



IMPACT SERIES 14

AI GETS PHYSICAL

AI-powered humanoid robots are set to augment the workforce.



Foreword

Welcome to the fourteenth Impact Series report from the Research team at Barclays, in which we examine a new frontier in Artificial Intelligence: AI-powered robots in human form. Seismic shifts in technology are bringing both risk and opportunities, transforming globally important sectors from defence to healthcare, manufacturing and hospitality.

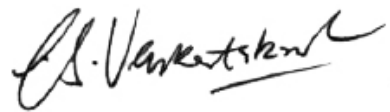
14 January 2026

When I entered the labour force more than thirty years ago, the idea of a humanoid co-worker was the stuff of science fiction. Even just a few years ago, the most cutting-edge machines seemed clumsy and limited in utility. But rapid advances in three areas – we call them the “three Bs” – are now changing the picture. Robots’ “Brains” have been much improved by cognitive AI models. Lightweight Batteries can now store much more power, more cheaply. Then comes the Brawn – the mechanical systems that convert energy into motion, enabling precise steps, grips and twists.

Humanoids could soon be taking on some of the tough, repetitive jobs that everyone needs, but few want to actually carry out. Such a transition could ease demographic headwinds in many developed countries that threaten growth and productivity, while reshaping cost structures in labour-intensive industries. Human beings, spared from the most

mundane or physically demanding tasks, could focus on higher-value activities. This shift is unlikely to be uniform everywhere and will depend, for example, on demographics and economic need, particularly in developing countries.

AI software has become deeply embedded in organisations, including our own, very quickly. It is a good time to consider what the next stage – physical AI – could mean for the future of work. We hope this report provides a useful contribution to an important debate.



C.S. Venkatakrishnan
Group Chief Executive Officer
Barclays



AI's next frontier is physical: humanoid robots – robots in human form – are stepping out of the lab and into the real world. They could take on the tough, repetitive jobs humans increasingly avoid in manufacturing, agriculture and healthcare – working alongside people and augmenting the workforce.

- **Humanoid robots: where physical power meets cognitive intelligence.** Automation began with machines doing the heavy lifting. Then came AI, adding a layer of intelligence. Now, humanoids fuse both – merging strength with smarts, as they step into a new era.
- **Demographic headwinds are creating demand.** Ageing populations, labour shortages, and shifting worker preferences are opening the door for humanoids to take on essential – yet often undesirable – roles in manufacturing, logistics, agriculture, healthcare, and hospitality. While many tasks in these sectors have been automated, others remain challenging because they require human-like interaction and dexterity – precisely where humanoids excel.
- **The “three Bs” – brains, brawn and batteries – underpin the humanoid supply chain.** Advances across all three have cut unit costs by 30x in the past decade. Of these, brawn – the robotic “muscles” converting energy into motion – is the centrepiece, driving roughly 50% of production costs. Powered by **actuators**, these muscles enable every step, grip, and twist. As production scales, actuator systems could be a critical bottleneck.
- **Humanoids are cars in miniature.** Each robot packs thousands of high-precision components and complex supply chains – mirroring the complexity of automotive manufacturing. This overlap means decades of expertise in actuators, gears, and bearings can be repurposed for humanoid production, creating a natural advantage for regions with strong automotive and industrial DNA – like Europe.
- **Physical AI could be the next big investment theme.** The humanoid market, just \$2-3 billion today, could surge to \$200 billion by 2035, according to our estimates. This shift creates a powerful tailwind for industrials – bringing actuator makers, automation leaders, and precision component suppliers into focus – companies largely absent from AI's software-driven first wave.

Will humanoids be the growth engine of the 21st century?

In 1961, General Motors introduced Unimate, the world's first industrial robot, to weld car parts on its assembly line – a moment that marked the beginning of large-scale physical automation. Fast forward to 2017, when Google's AlphaGo defeated Ke Jie in the strategy game Go, showcasing the power of cognitive AI. Now, entering 2026, the frontier is shifting again: BMW's Spartanburg plant put humanoid robots from Figure on the line, loading sheet-metal parts into welding fixtures and helping roll out 30K vehicles.¹ No longer confined to research labs, these machines are stepping – quite literally – into the real world. AI is getting physical.

Why it matters: demographics and demand

Three demographic shifts are reshaping the global economy. First, ageing populations are creating headwinds as the share of people aged 65 and above rises from 10% today to 16% by 2050, while working-age cohorts stagnate. Second, urbanization is accelerating, with nearly 70% of the world projected to live in cities by midcentury – concentrating talent in urban hubs and leaving sectors like manufacturing, agriculture, and elder care in rural areas short of labour. Third, job preferences are changing – not always due to pay, but because they are repetitive and monotonous, physically demanding, hazardous, or simply less appealing than other opportunities.

The result is a growing mismatch between economic demand and workforce supply in critical sectors such as manufacturing and logistics, agriculture, and healthcare. While many roles in these industries are already automated with industrial robots, others that are less structured and require human-like dexterity and adaptability have resisted automation. Our view: humanoid robots – machines designed to resemble and interact like humans – are positioned to bridge that gap.

Why humanoids?

Humanoid robots need a human form because the world is built for humans. From navigating narrow hallways to climbing stairs or opening doors, a human-like shape lets robots operate in existing environments without costly redesigns. Take a sorting centre: it is a fast-moving environment with packages

of varying sizes, conveyor belts, shelving, and equipment designed for human reach and movement. A wheeled or single-purpose robot might excel at one task – like scanning labels – but would need reprogramming to stack boxes or clear jams. A humanoid robot, equipped with arms, legs, and visual reasoning, can adapt and perform multiple tasks safely.

The big shift

Until recently, however, these robots lacked one critical skill: understanding context. In the real world, that meant mistakes as absurd as watering a person instead of a plant, struggling to identify transparent surfaces, or confusing solid objects with flimsy ones. Thanks to advances in AI, humanoids are now learning to read situations and act accordingly – making them far more scalable than fleets of specialized machines. The second challenge was hardware: production costs remained the biggest barrier to mass adoption.

The “three Bs”: brains, brawn and batteries

Our analysis highlights what we call the “three Bs” – brains, brawn and batteries – as the core technologies behind humanoids and their main cost drivers. Brains cover cognitive AI models and physical compute; brawn refers to the mechanical systems for movement, dominated by actuator components including electric motors, gears, bearings, and sensors; and batteries supply energy. Of these, the brawn – the “muscles” converting energy into motion – is the centrepiece of the humanoid architecture, accounting for about 50% of total humanoid production costs, according to our estimates.

¹ See press release: [F.02 Contributed to the Production of 30,000 Cars at BMW](#).

Cost collapse and investors taking notice

Breakthroughs across the “three Bs” have driven a dramatic collapse in costs – by our estimates, a 30x drop over the past decade, from around \$3mn to roughly \$100K per unit. As humanoids are becoming more economically viable, investors are taking notice. Venture capital funding hit \$8.8 billion in Q2 2025, up 15x since 2017. The valuation of Figure, one of the leading start-ups in the humanoids space, jumped from \$2.6 billion to \$39 billion in just seven months, and mentions of “humanoid” robots have risen 10x since 2020, based on our analysis of a wide range of reports, newspaper articles, and company filings. Meanwhile, 21 new humanoid models were unveiled in 2025, signalling a wave of innovation across industries. Today, China leads in production and deployment, but that could change.

Europe’s winning “hand”?

Cars transformed the 20th century. Could humanoids do the same for the 21st? Humanoids will not become mainstream tomorrow – but if and when they do, actuators likely will define the supply chain. Many robotics firms are software-first and lack the deep manufacturing experience needed for large-scale production. Yet actuators are not exclusive to humanoids – they have deep roots in other industries, notably automotive, as well as industrial automation and aerospace. Europe’s car-making DNA in precision mechanics could prove to be its strongest advantage.

Our analysis of UN Comtrade data shows Europe leads global actuator supply with a 34% share, ahead of China (26%), Mexico (9%), Japan (5%), and the US (5%). Given Europe’s highly integrated supply chains, viewing it collectively makes sense – but one fact stands out: Germany alone accounts for about one-third of Europe’s actuator supply.

A humanoid is essentially a car in miniature: thousands of fine-tolerance parts, multi-tiered supply chains, and advanced forming technologies. It faces the same mechanical challenges: complexity, precision and scale. Take robotic hands: among the most intricate and costly components of humanoids, they account for nearly a fifth of the total build cost and require dozens of specialised bearings and gears, much like the intricate systems in cars. This overlap extends across actuators, motors, and sensors, meaning decades of automotive expertise and manufacturing capacity, especially in Europe, could be repurposed for humanoid production.

No wonder partnerships are booming – BMW with Figure AI, Schaeffler with Nera, Hyundai with Boston Dynamics, and more, driven by growth ambitions and the promise of humanoids on factory floors. Another advantage: car factories are an excellent proving ground for humanoids – allowing engineers to capture real-world insights and feed those takeaways into future models.

The next big investment theme: physical AI

The humanoid robot market, just \$2-3 billion today, could surge to \$40 billion by 2035 – or even hit \$200 billion in the most optimistic scenarios. This growth underpins three major investment opportunities.

First, scaling physical AI and humanoid production creates a powerful tailwind for industrials – from actuator and component manufacturers, to leaders in industrial automation – bringing into focus companies that were largely absent from the first wave of AI. The initial wave centred on software, dominated by hyperscalers and semiconductor giants building models and data infrastructure. The second wave is different: it is about physical AI – hardware, precision mechanics, and advanced manufacturing – an entirely new group of players stepping into the spotlight.

Second, defence tech is a major spillover from humanoid robotics. Many components are dual-use – the same magnet motor that moves a humanoid arm can power a submarine or steer a missile. As global defence spending surges toward \$6.6 trillion by 2035,² this convergence should only deepen, with progress in one domain accelerating the other: advancements in humanoids feeding into defence technologies like drones and autonomous ground vehicles – and vice versa.

Third, future physical AI technologies share a common dependency: critical minerals and rare earths. Over 90% of [magnetic rare earths](#) and a significant share of other critical minerals come from China, creating strategic vulnerabilities as demand surges with adoption, echoing the role of oil during the automotive boom. Diversifying supply chains and building a mine-to-device ecosystem likely will be essential – creating a structural tailwind for mining companies outside of China.

The future of work

Humanoid robots could help augment the labour force in ageing economies, easing demographic headwinds that weigh on growth and productivity. By taking on repetitive,

² See: [United Nations](#).

physically demanding, or hazardous tasks across sectors like manufacturing, logistics, agriculture, and healthcare – jobs poorly suited to traditional automation – they offer a structural response to labour shortages. Their advantage lies in near-continuous operation: even at half human efficiency, a humanoid can deliver 25% more output per day, and at parity, up to 150% more, according to our analysis.

Over time, adoption could reshape cost structures in labour-intensive industries, shifting from rising operating expenses (opex, which includes wages, benefits, compliance) to upfront capital expenditure (capex) on robotics – creating opportunities for long-term margin expansion and altering macroeconomic dynamics around employment and capital allocation. Importantly, robots should complement – not replace – humans. By taking on routine and physically demanding tasks, humanoids could allow human workers to focus on higher-value tasks, improving productivity and service quality. Integrating humanoids into the workforce would also create greater demand for oversight and control roles – responsibilities that we believe would remain firmly under human authority.

Policy choices ahead

Humanoid robots hold significant potential to boost productivity and support ageing societies in developed nations. In contrast, developing nations – with younger workforces and lower labour costs – are likely to follow a different path: near-term adoption may remain limited and there are potential economic headwinds. However, opportunities exist to leapfrog mature technologies, enhance workplace safety, and expand access to healthcare where infrastructure constraints persist.

The overarching challenge is to ensure that the benefits of humanoid robotics are shared equitably – both within nations and globally – so these innovations advance human welfare rather than deepen divides. Governments may need to explore options such as reskilling programs, updated regulations, or new funding mechanisms to support social welfare.





The essential jobs everyone needs but few want – humanoids can do them

Three powerful demographic forces – population ageing, rapid urbanisation and evolving work preferences – are rewriting the rules of the global economy. This is creating pressure points in key sectors such as manufacturing, logistics, agriculture, and healthcare, where economic demand is rising faster than labour supply. Our view: humanoid robots – machines designed to resemble and interact like humans – are poised to close that gap over time.

1: Populations are ageing

According to UN projections, those aged 65 and above will rise from 0.9 billion today to 1.6 billion by 2050, growing from 10% to 16% of the population, and surpassing 30% in some regions like Europe, Japan, and China (Figure 1). This shift matters because fewer workers must support more retirees, straining pensions and healthcare, while labour shortages could hit productivity. [Population ageing](#) is not just a demographic trend – it could become a macroeconomic headwind that reshapes fiscal policy, labour markets, and social infrastructure.

2: Urbanisation is accelerating

The UN projects that by 2050, nearly 70% of the world's population will live in urban areas (Figure 2). In certain developed economies this figure is projected to exceed 80%: to 84% in Europe, 89% in the US, 90% in the UK and 95% in Japan – up from today's levels of 76%, 84%, 85% and 92%,

respectively. While this fuels growth and innovation, it also creates mismatches: manufacturing, agriculture, and elder care remain rural, while labour and resources concentrate in cities. This gap could strain supply chains, complicate rural healthcare, and raise logistics costs for essential goods.³

3: Work preferences are changing

Some occupations are “ageing” faster than others, as younger generations (Gen Z and Gen Alpha) that enter the workforce are less inclined to take them on. These roles are often repetitive, physically demanding, hazardous or simply less appealing compared to other opportunities – and not necessarily because of pay. With a shrinking population and shifting work preferences, certain jobs and industries will face much tighter labour markets than others.

³ World Health Organization. [Health and care workforce in Europe: time to act](#), p.2.

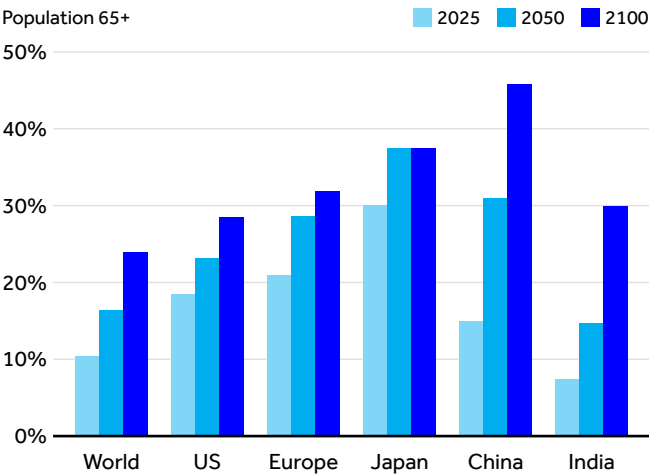
Demographic headwinds

Taken together, these three demographic shifts are reshaping labour supply-demand dynamics, amplifying shortage risks especially in agriculture, healthcare and elderly care, and manufacturing and logistics.

Agriculture is among the sectors with the oldest labour force (Figure 3). In Europe, agricultural workers are on average three years older than the general workforce; in Japan, the gap is about ten years. The US stands out slightly, largely due to an

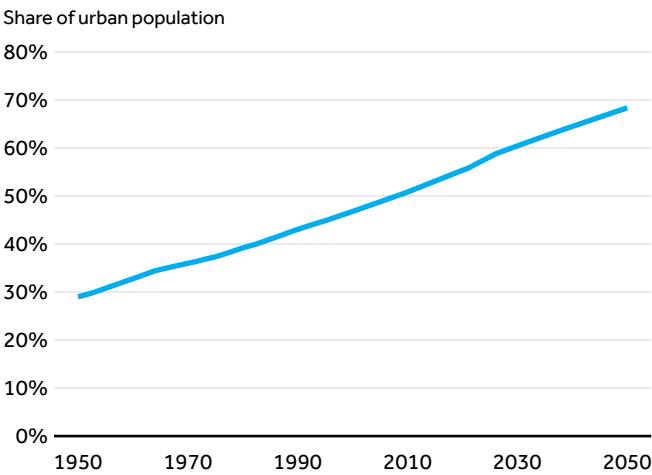
immigration wave between 2021 and 2024. At the same time, the agricultural labour force is shrinking significantly: between 2010 and 2024, it declined by 37% in Europe, 23% in Japan, and 17% in the US (Figure 4). While part of this decline stems from technological substitution, such as mechanisation and automation, reducing the need for manual labour, the fact that agriculture has one of the oldest workforces indicates that natural attrition through retirement is a major driver, independent of automation.

FIGURE 1
The share of people aged 65 and over is set to keep rising across major economies



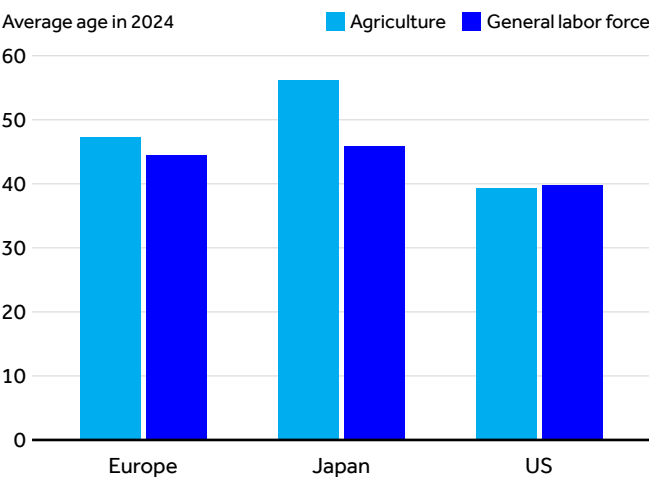
Source: United Nations' World Population Prospects 2024, Barclays Research

FIGURE 2
By 2050, two-thirds of humanity likely will live in cities



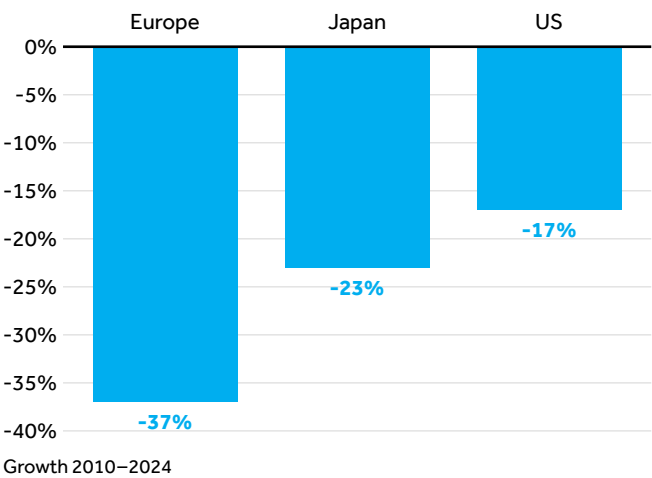
Source: United Nations' World Urbanization Prospects 2025, Barclays Research

FIGURE 3
Agriculture is among the sectors with the oldest labour force...



Source: American Community Survey, Eurostat, Statistics Bureau of Japan, Barclays Research

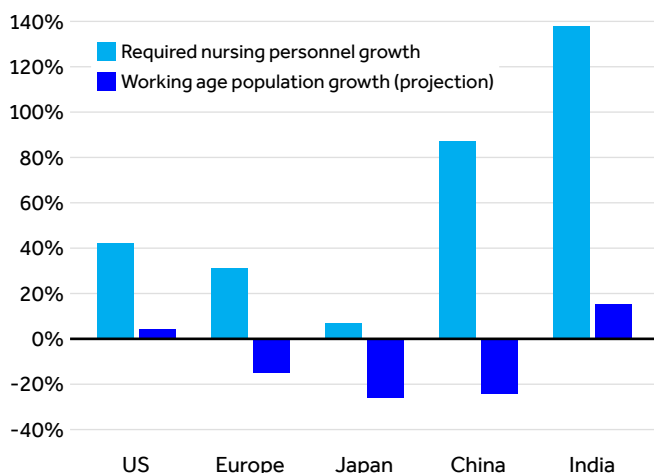
FIGURE 4
...and its labour force is rapidly shrinking



Source: American Community Survey, Eurostat, Statistics Bureau of Japan, Barclays Research

FIGURE 5

Demographic headwinds: why healthcare could face labour shortages in the future



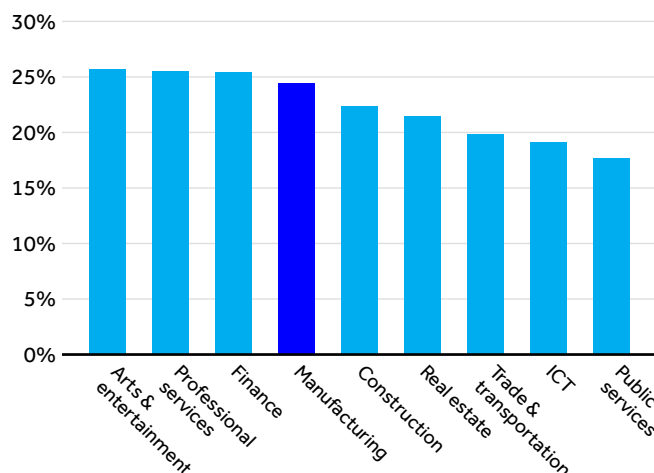
Note: Projected growth in nursing staff (2025–2050) required to maintain today's ratio to population aged 65+.
Source: United Nations' World Population Prospects 2024, World Health Organization, Barclays Research

Healthcare and elderly care faces similar challenges. To keep today's ratio of nursing staff to people aged 65 and above by 2050, demographic trends imply that nursing personnel would need to grow by about 40% in the US and 30% in Europe (Figure 5). In younger economies like China and India, the required increase exceeds 100%. Meanwhile, the working-age population is shrinking or growing at a rate, which in our view, may not be fast enough to satisfy the rising demand. While integrating robots into elderly care is complex, there is the potential for them to take on basic tasks such as patient lifting or administering medication, freeing up capacity for more specialised, human-led care.

The manufacturing industry is facing a perfect storm: the retirement of the baby boomers alongside economic and societal factors that have contributed to labour shortages. OECD analysis shows that between 2010 and 2022, labour tightness – measured as vacancies per unemployed worker – increased across sectors (Figure 6). However, we think that the underlying dynamics differ. In arts, finance, and professional services, tightness largely reflects temporary mismatches: workers want these jobs, but skills and timing lag. Manufacturing, by contrast, faces a more structural supply elasticity problem. Physical demands, limited flexibility, geographic constraints, and perceived low prestige mean that additional openings do not attract proportionate interest.

FIGURE 6

Increase in labour market tightness by sector in OECD countries, 2010–2022



Note: We show the tightness increase between 2010–2022. Tightness is defined as the number of job vacancies per unemployed person.
Source: OECD, Barclays Research

In the US, more than 25% of manufacturing workers are over 55, this means that the retirement of baby boomers will intensify pressure on a sector already struggling to appeal to younger generations, who favour roles in industries seen as more flexible or less physically demanding.⁴ A survey by Soter Analytics found that only 14% of Gen Z would consider a career in industrial work, making tightness in manufacturing fundamentally harder to resolve.⁵

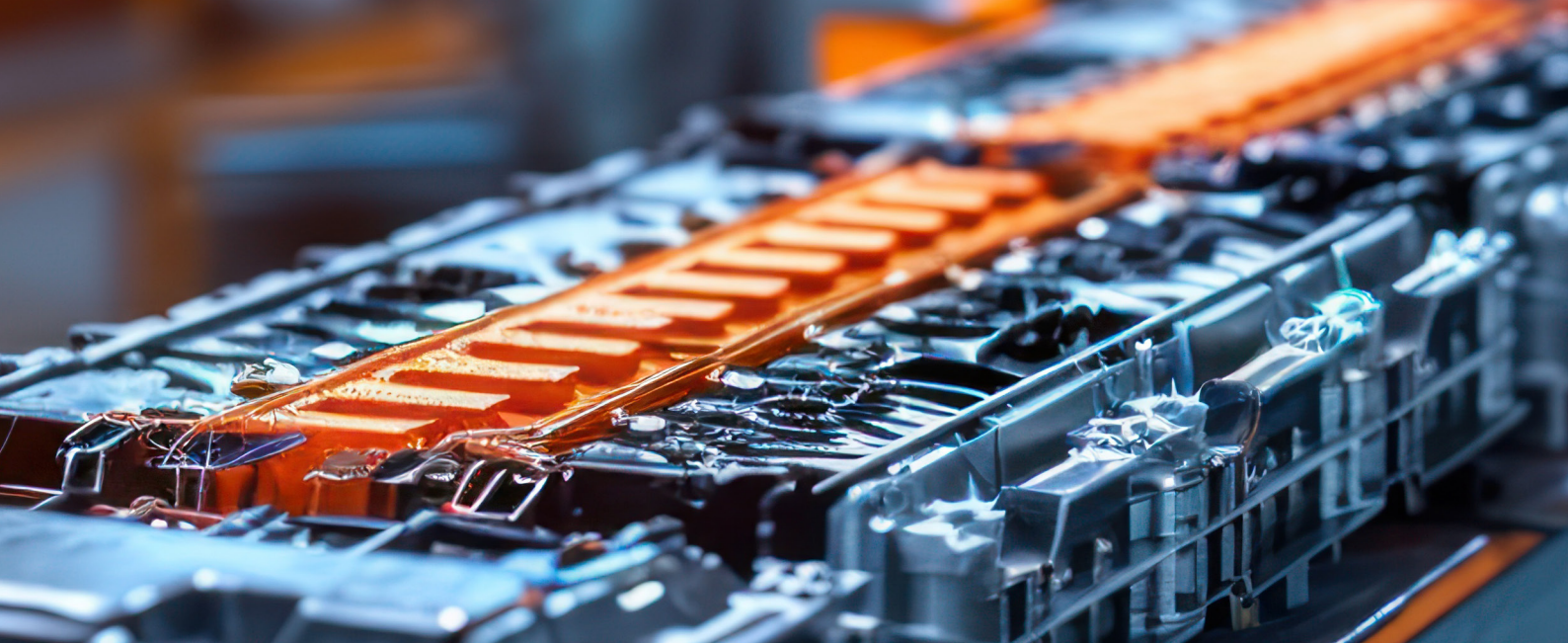
Humanoid robots could offset some of these challenges

Historically, many roles in these industries have resisted automation due to the need for human-like dexterity and adaptability. We believe that humanoid robots, designed to mimic human form and behaviour, can bridge this gap by performing tasks beyond the reach of traditional industrial robots, sustaining productivity in labour-intensive sectors. Combined with existing robots for rigid, precise work, the flexibility of humanoids can unlock additional automation potential.

While humanoids might not be on their way to work in factories, stores and hospitals just yet, they are quickly evolving. The next section outlines the three main components of the humanoids' architecture.

⁴ See: [OECD Economic Outlook](#) and [Manufacturing today](#).

⁵ See World Economic Forum report: [From tech to sustainability: Why the industrial sector is calling on Gen Z and Gen Alpha to join its workforce](#).



The three Bs of the humanoid supply chain: brains, brawn, batteries

The three Bs are the foundational building blocks of the global humanoid supply chain and the biggest investment opportunity (Figure 7).

While much of the current discourse focuses on the “brains” or “batteries” of humanoid robots, we believe the real engineering challenge lies in the “brawn” – specifically, replicating the dexterity and autonomy of the human body. Our analysis shows

that “brawn”, dominated by actuators, accounts for roughly 50% of total humanoid production costs, compared with 35% for “brains” and 15% for “batteries.” In our view, mechanical systems and actuators represent the most critical link in the humanoid supply chain, as they would ultimately determine whether humanoids can be manufactured at scale and deployed safely on the factory floor as well as in the household.

FIGURE 7
The three Bs of the humanoid robot architecture

BRAINS (35%)							
AI stack		Compute		Sensory systems			
Base models	Software	CPU & GPU	Memory	Radar & Lidar	Microphones	Cameras	Proximity sensors
BRAWN (50%)							
Actuator systems							
Complete actuators	Motors	Gears/reducers	Brakes	Bearings	Screws	Rare earths and magnets	
Control, power and thermal units	Functional core			Structural frame			
	Force & torque sensors		Analog semis		Castings, wiring, cables, fasteners, alloys		
BATTERIES (15%)							
Battery cells	Battery management		Thermal management		Battery pack structure		Power distribution

Source: Barclays Research

The brains

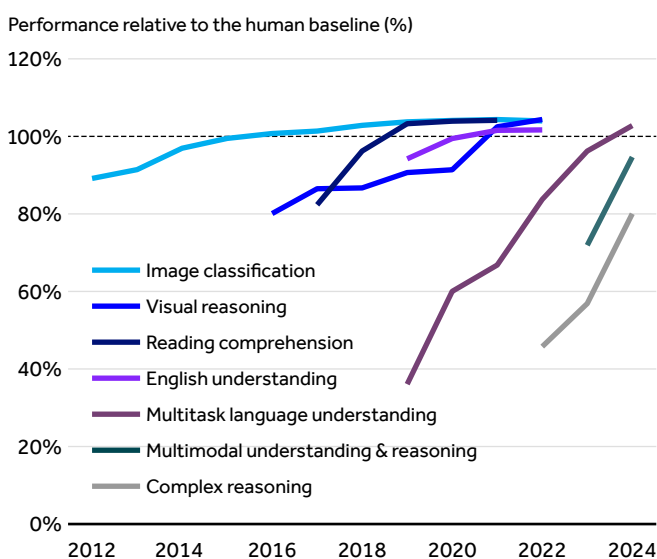
Humanoids leverage advances in AI models for vision, motion, and language, enabling them to perceive, reason, plan, and interact effectively with their surroundings. These capabilities are essential for operating in unstructured environments: real-world contexts that are complex and constantly changing, where rigid automation or preprogrammed tasks fall short.

Recent breakthroughs in perception, reasoning, and hierarchical control have dramatically improved humanoids' ability to interpret context, adapt and act intelligently in dynamic environments. Figure 8 highlights how capabilities in key domains related to humanoid deployment have steadily advanced to meet – and in some cases exceed – human-level benchmarks (index = 100%). For example, image classification surpassed human performance in 2016, while visual reasoning beat humans in 2021. Most recently, multitask language understanding surpassed human ability in 2024. The latter models enable a single AI system to perform multiple operations, rather than relying on separate models for each task.

However, there is still room for improvement, particularly in multimodal understanding and reasoning. This capability, which enables humanoids to combine vision, sound, and touch to fully grasp their environment, remains below human benchmarks. Complex reasoning – the ability to handle multistep logic, connect abstract concepts, and draw conclusions across different contexts – is another area still lagging behind human performance, though the gap is closing fast.

FIGURE 8

Gains in AI perception and reasoning have paved the way for real world humanoid deployment



Note: We show performance of AI models relevant to humanoid deployment.

Source: AI Index, Barclays Research

Models typically use a hybrid approach: local compute for real-time tasks and cloud connectivity for updates and maintenance. Equipped with mini supercomputers like NVIDIA Jetson, which combines GPU and CPU capabilities, humanoids can run AI models locally for vision processing, speech recognition, and motion planning with low latency, ensuring smooth and responsive performance. These capabilities rely on inputs from high-resolution cameras and diverse sensors, which provide visual, tactile, and spatial data essential for accurate perception and safe interaction.

The brawn

However, even the smartest AI needs a way to translate decisions into physical motion – and that is where actuators come in. Actuators are integrated assemblies that typically include an electric motor, gears to adjust speed and force, and bearings to ensure smooth rotation, along with housing and sometimes sensors for feedback. Designing actuators that are powerful yet lightweight, precise yet affordable is one of the toughest challenges.

An actuator is essentially the robot's "muscle" – the component that turns energy into motion, enabling it to bend an arm, walk, or turn its head. For humans, the brain sends a signal, muscles contract, and the arm moves. For robots, an AI model sends a signal, and the actuator moves the part. Actuators come in different forms (eg, rotary in shoulder and hip joints, and linear in legs and lifting mechanism), but they all share the same purpose: to make the robot move in a controlled, lifelike way.

Today's humanoid robots have far fewer joints than humans, creating a gap between their movements and those they aim to replicate – a limitation that significantly reduces their utility. Humans move with grace and efficiency because the body is a symphony of compliant joints, flexible spines, and spring-like tendons. By contrast, a humanoid robot is a rigid assembly of metal and motors, linked by joints which still have limited degrees of freedom.

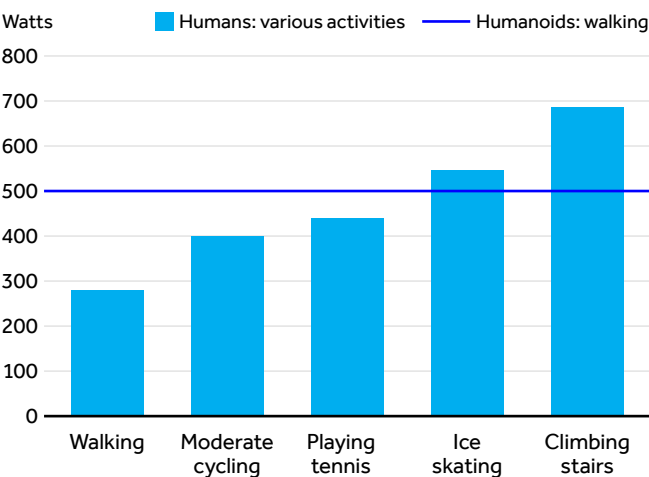
Advancements in actuators technology have improved the precision, response times, and torque control and are making humanoids motion more natural and human-like. Torque is the twisting force that makes things rotate, similar to how pressing the accelerator in a car controls how strongly the wheels turn: better torque control means smoother, steadier movements for the robot. These improvements translate into greater dexterity for tasks such as grasping delicate objects or performing complex manipulations. In addition, enhanced actuator systems improve safety through constant movement corrections that maintain stability and balance, reducing the risk of falls or unintended motions and making humanoids more reliable in environments shared with humans.

The batteries

Humanoids rely on lithium-ion batteries for power, using lightweight, high-capacity systems built from lithium and other critical minerals like cobalt, nickel, and manganese. Actuator systems consume the bulk of this energy, as every movement requires power to overcome weight and inertia. Staying upright alone demands millions of rapid, energy-intensive corrections each second. Current battery technology provides run times of about 4–5 hours, but high-intensity motion can drain power even faster.

FIGURE 9

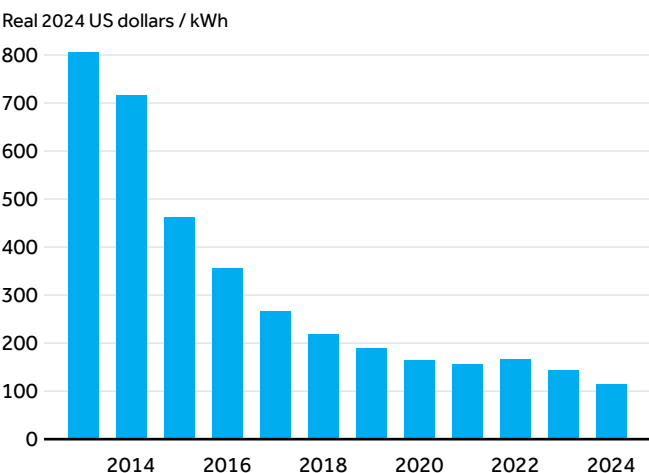
Humanoids' energy footprint



Source: AlphaSense, Douglas College Physics 1104 Custom Textbook, Barclays Research

FIGURE 10

Lithium-ion battery prices have dropped significantly over the past decade



Source: Bloomberg BNEF (Bloomberg New Energy Finance), Barclays Research

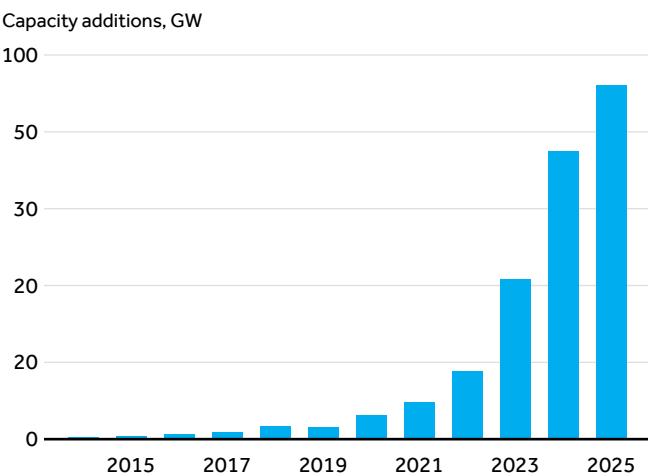
As a result, humanoids' energy efficiency still significantly lags behind that of humans. For example, the more advanced humanoids, such as Tesla's Optimus robot, are estimated to operate at about 500 watts on average while walking. By comparison, a human performing the same task is 40-45% more energy-efficient; in fact, only during more physically intense activities does the average human exertion reach 500 watts (Figure 9).

Advances in battery tech are laying the groundwork for the next generation of humanoids. Lithium-ion battery prices have dropped nearly 8x over the past decade, reaching about \$100 per kWh by 2024 (Figure 10). At the same time, global energy storage capacity has expanded rapidly, with annual additions breaking records year after year (Figure 11). Together, these developments have made integrating batteries into humanoids increasingly practical.

Considering factors such as component count, complexity, and unit costs, we estimate that brawn components represent around 50% of the bill of materials, brains account for 35%, and batteries make up the remaining 15%. While these figures are indicative, we believe they are reasonable. Brawn dominates because humanoids require numerous high-precision electromechanical components with tight tolerances, making them costly to manufacture and assemble. In contrast, brains are concentrated in fewer units – such as processors, cameras, and control bands – which benefit from semiconductor economies of scale, reducing their relative share. Batteries are not a major component because energy storage requirements for humanoids are modest compared to electric vehicles, limiting both size and cost.

FIGURE 11

Global battery storage availability continues to rise



Note: Capacity additions measure new storage installed globally each year, not the total stock.

Source: International Energy Agency, Bloomberg BNEF (Bloomberg New Energy Finance), Barclays Research

Humanoid robots are no longer sci-fi

Breakthroughs across the “three Bs” have driven a dramatic collapse in costs. While precise figures are not publicly disclosed, our estimates – based on a synthesis of company reports, industry articles, white papers, earnings transcripts, and expert commentary – suggest a dramatic 30-fold reduction over the past decade: from approximately \$3mn to around \$100K per unit (Figure 12), with a wide range between low-end configurations and cutting-edge models.

The market is taking notice

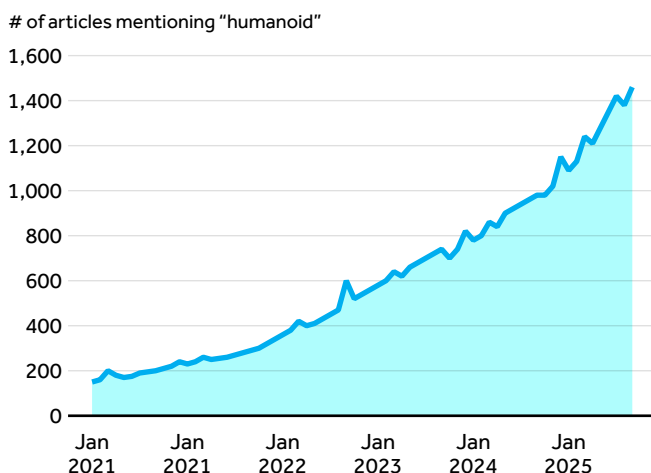
Media and market attention are accelerating: mentions of the word “humanoid” in news and reports have grown nearly 10x, from 150 in January 2020 to 1,500 in September 2025, reflecting a sharp rise in public and industry engagement (Figure 13).

At the same time, innovation is accelerating: 21 new humanoid robot models were introduced in 2025 alone, compared to just three in 2022 and none in 2021, reflecting a wave of innovation across industrial, domestic, and service applications (Figure 14). These latest models are not only more numerous but also markedly more advanced, featuring significant improvements in functionality and design. Notably, close to 70% of the new models unveiled in 2025 originate from Chinese companies, highlighting China’s growing dominance in this space.

Figure 15 shows the top ten humanoid robot developers – defined as companies that have unveiled at least two models – listed in alphabetical order. The majority are private, with notable exceptions such as Tesla, Toyota, UBTECH and XPENG.

FIGURE 13

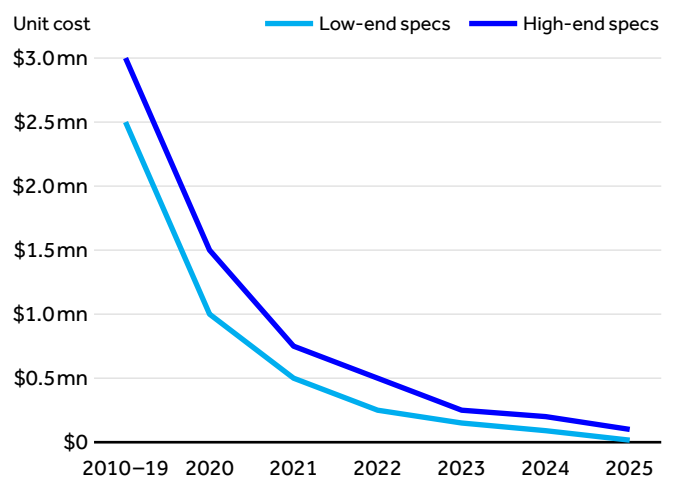
The market is “talking” a lot about humanoids



Source: AlphaSense, Barclays Research

FIGURE 12

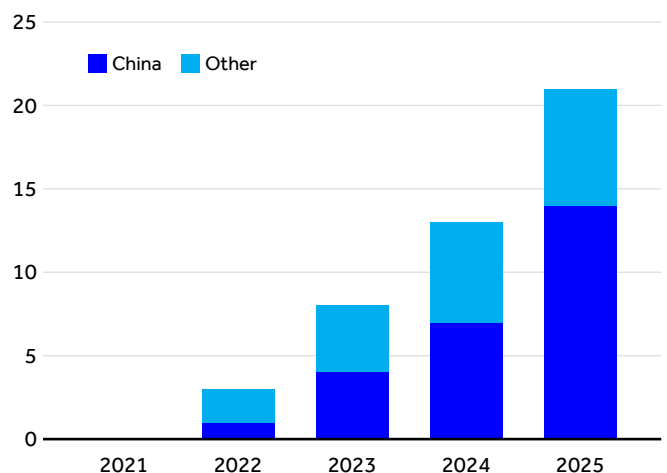
Humanoids’ unit production costs have plunged 30x over the past decade



Source: AlphaSense, Company reports, Barclays Research

FIGURE 14

Number of new humanoid robots unveiled



Note: Based on company reports and announcements. Includes bipedal humanoid robots.

Source: Company Reports, Barclays Research



Humanoids' unit production costs have plunged 30x over the past decade.

China makes the most humanoids

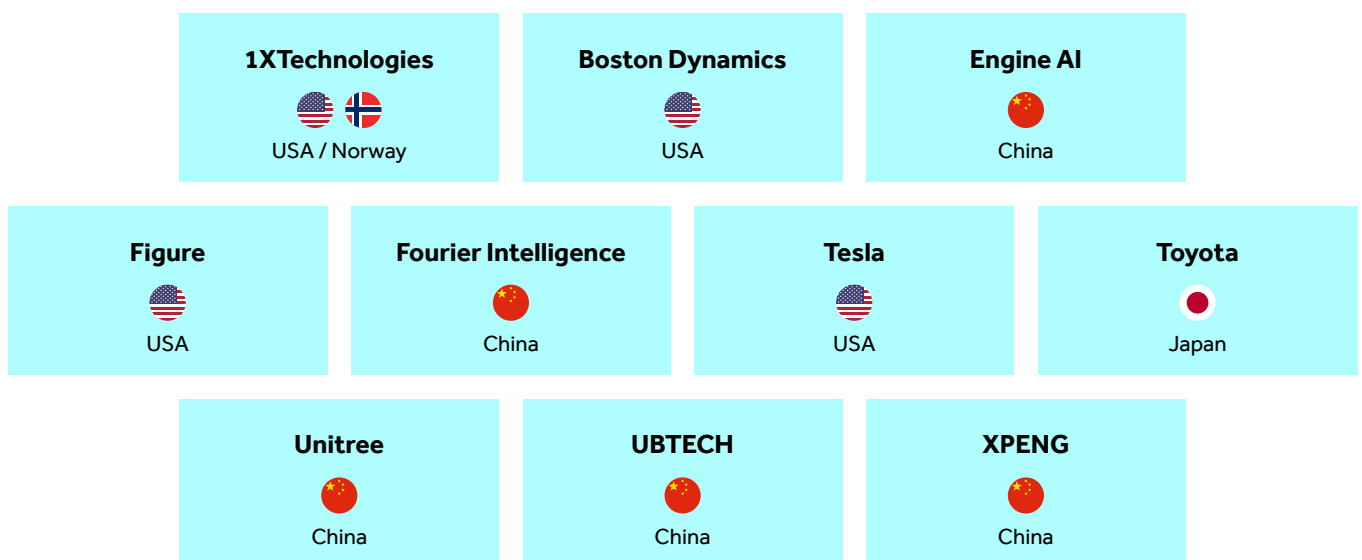
China is currently the global leader in humanoid robotics, driven by affordability, government backing, self-sufficiency in critical raw materials like rare earths and tight vertical integration positions. Unitree has disrupted pricing with a steep trajectory: its humanoid robot model, H1, launched in 2024 at \$90K, followed by G1 at \$16K, and now R1 at just \$5.9K – making humanoids dramatically more accessible. In November 2025, UBTECH marked a major milestone with the first mass

production of full-size industrial humanoids, the Walker S2, targeting 500 units in 2025 and scaling to 10K annually by 2027. This could signal the start of large-scale adoption and China's push to set the pace globally.

While China leads today, the real tipping point would come when humanoids move beyond prototypes and into mass production worldwide – and that is where Europe has a unique opportunity.

FIGURE 15

The top 10 humanoid developers



Note: Includes companies that have unveiled at least two humanoid robotic models. Companies are listed in alphabetical order.
Source: Barclays Research

Can Europe's automotive DNA win the robotics game?

Cars transformed the 20th century – could humanoids do the same for the 21st? Adoption will not happen overnight, but if and when scale comes, actuator systems likely will define the winners and losers in the supply chain. Europe's deep-rooted expertise in precision mechanics – honed through decades of industrial engineering, from automotive to advanced manufacturing – might lead to a leading position.

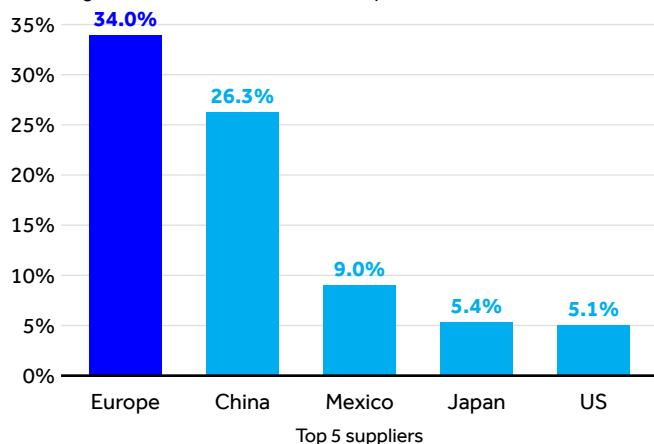
Our analysis of UN Comtrade – the world's most comprehensive trade database – shows Europe (EEA plus Switzerland) is the leading supplier of actuator systems and components (Figure 16). Europe accounts for 34% of global supply, ahead of China (26%), Mexico (9%), Japan (5%), and the US (5%). Because European supply chains are deeply interconnected – components often cross borders multiple times during production – it makes sense to view Europe collectively. But when we break down the flows, one fact stands out: Germany provides roughly one-third of Europe's supply of actuator components (Figure 17).

Europe's edge in mechanical systems stems from hosting a third of the world's current high-precision actuator manufacturers. Based on our analysis of the top 75 actuator firms worldwide (many of which are private), 23 are headquartered in Europe, with 13 in Germany. The US, Japan, and China follow as the other main hubs for actuator development (Figure 18).

FIGURE 16

Europe leads in the production of actuators: the lifeblood of humanoid robots

Share in global trade flows of actuator components



Note: 2024 data (latest available). Europe = EEA + Switzerland
Source: UN Comtrade, Barclays Research

A humanoid robot is a car in miniature

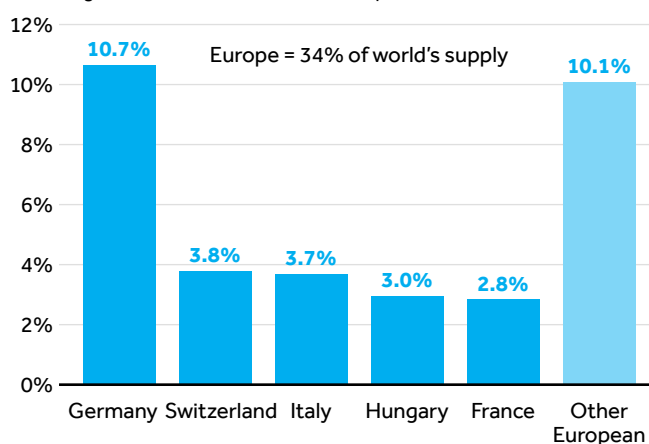
Humanoid production faces three core mechanical challenges: complexity, precision and scale. One potential strategy is deep vertical integration within the supply chain, but this is often impractical as our analysis shows that many robotics start-ups in the US and Europe are focused on AI and software rather than manufacturing, with Tesla being a notable exception. A more viable approach would be to form partnerships or acquire actuator specialists and industrial giants.

Building thousands of high-precision components at scale requires a multi-tiered supplier network, much like the automotive industry. Cars benefit from a mature supply chain, while the humanoid robot industry must create one from scratch. Yet they may not need to start entirely anew: we find that there is significant overlap in technology and infrastructure between the two sectors.

FIGURE 17

Germany anchors Europe's actuator edge

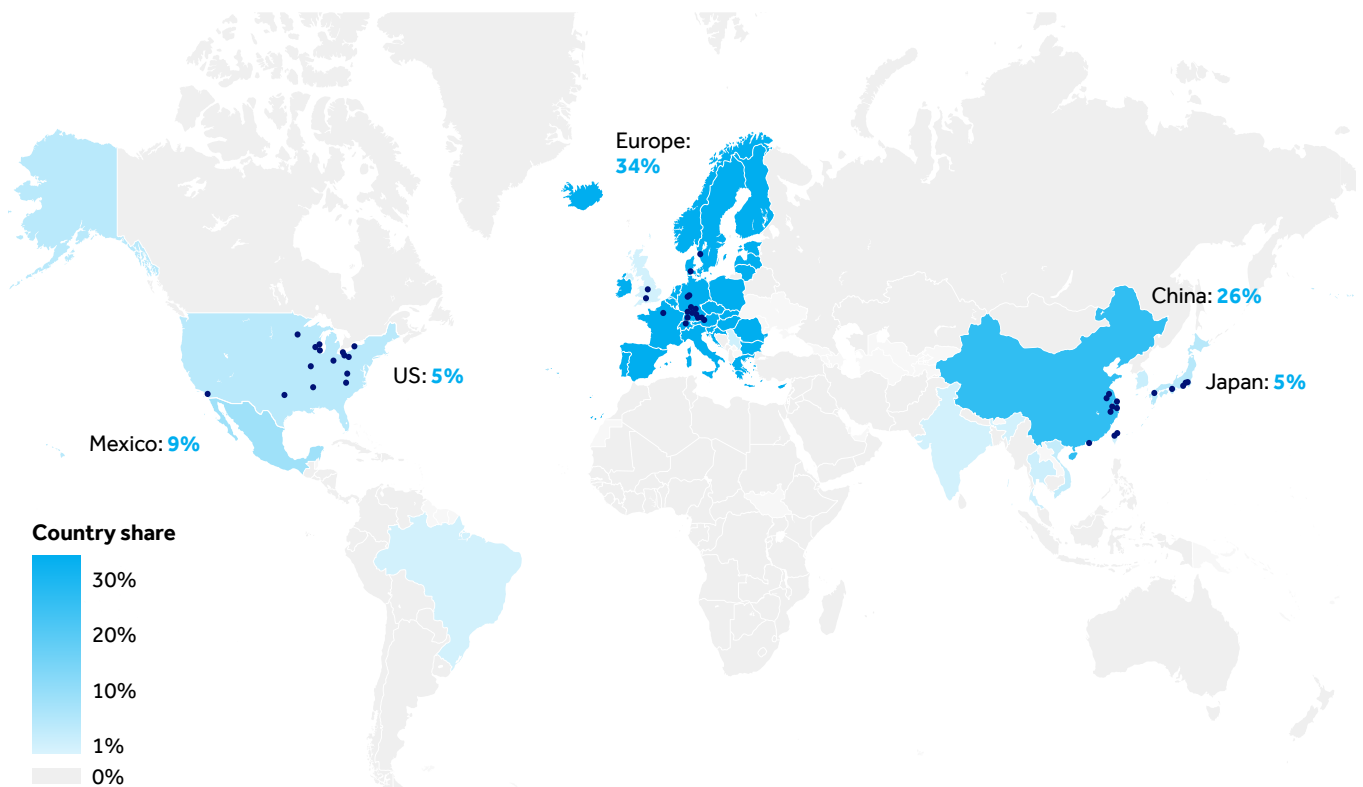
Share in global trade flows of actuator components



Note: 2024 data (latest available). Europe = EEA + Switzerland
Source: UN Comtrade, Barclays Research

FIGURE 18

Europe, China, the US and Japan produce the majority of high-precision actuators and host these companies' headquarters



Note: Shading represents the share of global actuator component production, based on UN Comtrade trade flow data. Blue dots mark the headquarters of 75 leading companies specializing in high-precision actuators.

Source: UN Comtrade, Company Reports, Barclays Research

The same types of bearings, gears and actuators found in cars can also be applied in humanoid robots. Take robotic hands, for example – they are the most complex and expensive components in humanoid production, accounting for nearly a fifth of total build cost. If each finger has three joints, and each joint requires a reduction gear supported by two ultra-small bearings, that amounts to 30 bearings per hand, or 60 per robot, but in reality, the number is likely higher. This density of parts mirrors the complexity of a car, which incorporates thousands of components sourced from specialized suppliers. Moreover, precision gears and bearings for humanoid finger articulation serve automotive applications such as electric seat adjustment systems or steering column tilt mechanisms. Likewise, high-torque, compact actuators designed for humanoid joints can be adapted for electric vehicle features like automated door openers or active aerodynamic flaps, where space and efficiency are critical.

Achieving lifelike finger articulation demands extremely tight tolerance – similar to those in an engine or transmission

system. Crucially, many of these precision parts, from bearings, motors, sensors, are already mass-produced for automotive applications. Decades of expertise in forming technologies for car parts can be repurposed for robotic joints, and as Figure 16 illustrates, much of this industrial capability is concentrated in Europe.

It is no surprise that many humanoid robot developers are forging partnerships with automotive manufacturers or suppliers. Examples include BMW with Figure AI, Schaeffler with Nera, Hyundai with Boston Dynamics, Tesla developing its own humanoid platform, and UBTECH collaborating with Geely's EV brand, Zeekr.

The automotive sector's interest is twofold: first, as a growth opportunity in an emerging technology; and second, as a practical solution for their own operations. Humanoids can be deployed on factory floors, working alongside humans in repetitive or physically demanding roles – the kind of jobs where shifting demographics should increasingly shape labour supply and demand in the years ahead.

The next big investment theme: physical AI

Humanoid robots are still a small market – just \$2-3 billion today – but growth prospects are enormous. Adoption is expected to remain gradual until 2030, when projections range from \$10-25 billion. By 2035, the gap widens dramatically: our baseline forecast reaches around \$40 billion, while under the most-optimistic scenarios, it climbs as high as \$200 billion (Figure 19).

Estimating the total addressable market (TAM) is challenging, and projecting its future even harder given the many moving parts. To address this, we aggregated a broad range of sources from AlphaSense – company reports, industry articles, white papers, earnings transcripts, and expert commentary – to build a forecast model across three scenarios: baseline, conservative, and optimistic.

This growth underpins three major investment opportunities: (1) new structural tailwinds for industrials, (2) spillover effects into adjacent sectors such as defence technology, and (3) rising demand for critical minerals essential to robotics and advanced manufacturing.

A tailwind for industrials

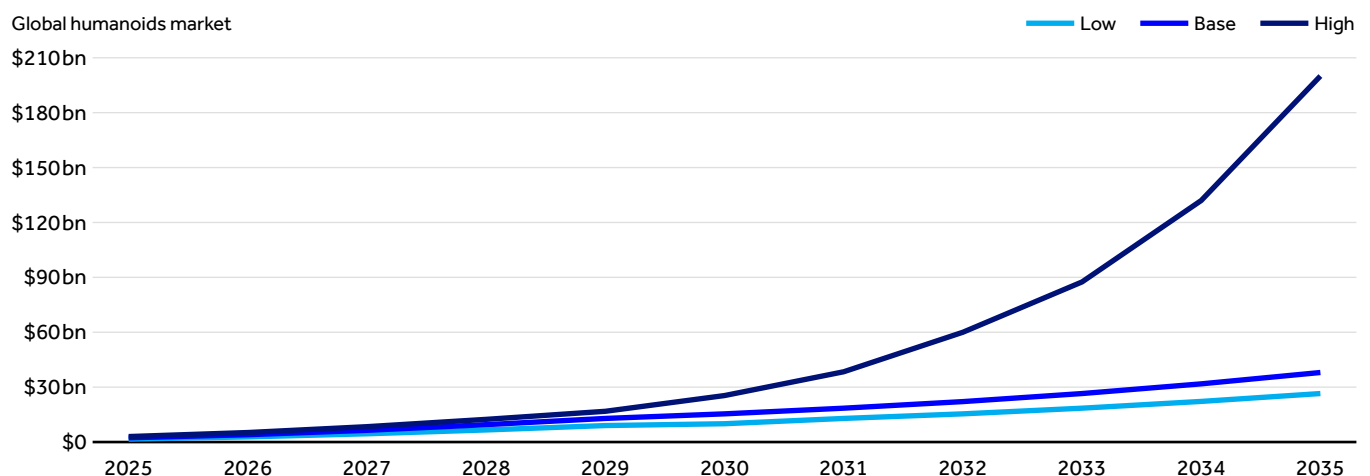
Scaling physical AI and humanoid production could unlock major upside for industrials – especially actuator system makers and component suppliers of motors, gears, and bearings, as well as industrial automation leaders and producers of sensors. Actuators rely on magnets made from rare earths, creating additional tailwinds for magnet and rare-

earth producers, particularly outside China. While hyperscalers and semiconductor firms should benefit as well, we think this opportunity is unique: many of these industrial players were not central to the first wave of AI growth, which focused on model building and software. A second wave – physical AI – brings an entirely new set of companies into the spotlight.

Investors can access the humanoid robotics/physical AI theme through several channels as companies seek financing for automation capex. Public firms may need to issue further debt to fund expansion, while much of the innovation in Europe resides in private, small- to mid-cap players, making venture and growth equity critical for early-stage exposure. For example, venture capital (VC) funding for robotics start-ups reached \$8.8 billion in Q2 2025, a 15-fold increase from just \$0.6 billion in 2017, according to PitchBook (Figure 20). One standout example is Figure AI, whose valuation ballooned from \$2.6 billion in February 2025 to \$39 billion by September 2025 (Figure 21). Finally, we expect M&A activity to accelerate over the next five years as industrials, robotics firms, and AI players pursue acquisitions and partnerships to secure technology, strengthen supply chains, and enhance integration capabilities.

FIGURE 19

The global humanoids market is set to grow over the next decade



Source: AlphaSense, Barclays Research

Spillovers into defence tech

The lines between humanoid robots, autonomous vehicles, and military systems are rapidly dissolving. These platforms increasingly share components, breakthroughs, and supply chains. Many robotics parts are inherently dual-use: the same high-performance magnet motor that animates a humanoid’s arm can propel an electric submarine or steer a missile’s fins. Progress in one domain accelerates the other – advancements in humanoids feed into defence technologies like drones, and vice versa.

This convergence is not theoretical. Boston Dynamics’ legged robots began as disaster-response and military projects; today, defence forces are actively developing unmanned systems – from bomb-disposal robots to future combat droids – that merge civilian and military robotics. A legged battlefield robot or an exoskeleton suit relies on the same core technologies as a commercial humanoid, ruggedized for combat.

The strategic implications are clear: global defence spending hit a record \$2.7 trillion in 2024 and is projected to reach \$4.4–

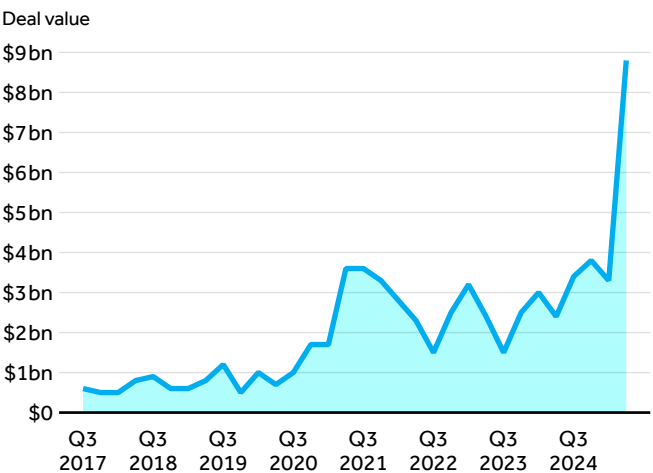
6.6 trillion by 2035 . As budgets surge, the overlap between physical AI and defence tech should only deepen, amplifying demand for critical materials and advanced components.

Demand for critical minerals

Future physical AI technologies – whether humanoid robots, defence systems, or autonomous drones – share a common dependency: critical minerals and rare earths. Our previous research shows that over 90% of magnetic rare earths are sourced from China, creating significant strategic vulnerabilities. China also dominates most of the critical minerals needed for batteries, which are included in the CLANC group: copper, lithium, aluminum, nickel, and cobalt.

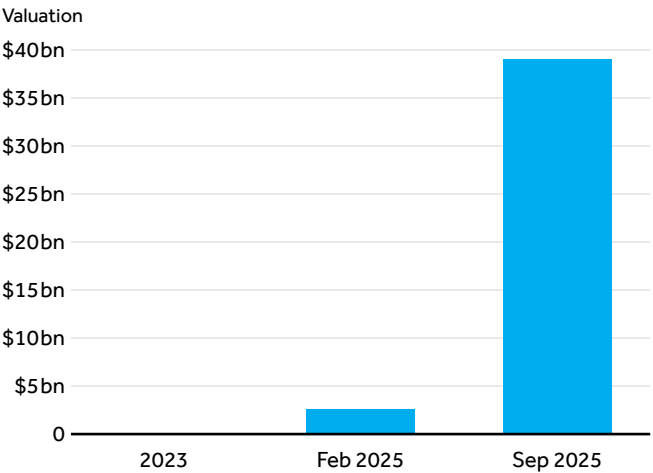
As physical AI adoption accelerates, demand for strategic minerals should surge – echoing the role of oil during the automotive boom. This dynamic also amplifies the imperative to diversify supply chains beyond China and develop a mine-to-device ecosystem, creating a structural tailwind for mining companies already positioned in this space.

FIGURE 20
Robotics VC deal activity is at an all-time high



Source: PitchBook, Barclays Research

FIGURE 21
Figure AI valuation has jumped



Note: Based on company press releases.
Source: Barclays Research



Work, reimagined: the macroeconomic shift ahead

Humanoid robots could be well suited to perform tasks in manufacturing and logistics, agriculture, healthcare and elder care and construction, helping to augment the labour force in ageing economies facing demographic headwinds.

Labour force augmentation and productivity

In Figure 22, we highlight roles with the greatest potential for humanoid robot adoption. These jobs share a common thread: they are repetitive, physically demanding, or hazardous, yet designed for human execution and poorly suited to automation by traditional industrial robots. Humanoid robots, with their human-like form and dexterity, can bridge this gap, enabling automation in environments previously inaccessible to conventional robotics.

While it is still uncertain whether humanoid robots can match human efficiency, their technical advantage lies in near-continuous operation. Unlike humans, a humanoid can run almost 24/7, with only short breaks for battery charging. For example, assume a human in a logistics facility delivers 8 productivity hours per day. A humanoid operating in 6-hour cycles (5 hours of work followed by 1 hour of recharge) allows for 20 work hours in a 24-hour period. If its efficiency is only half that of a human, that equates to 10 productivity hours per day, or 25% more output than a human. If the humanoid matches human efficiency, its output jumps to 20 productivity hours per day, or 150% more than a human worker.

In less-structured environments such as hospitals, the benefits

of humanoids lie in managing tasks beyond regular nursing hours. They can handle continuous deliveries of lab samples and supplies around the clock, disinfect rooms overnight to ensure they are sterile for patients in the morning, and provide uninterrupted patient monitoring.

Towards capex?

The integration of humanoid robots could reshape cost structures for labour-intensive businesses, shifting from a reliance on recurring opex such as wages, benefits, and compliance toward greater upfront capex on robotics. Unlike human wages, which typically rise with inflation and skill premiums, robot prices should depreciate over time, creating opportunities for long-term margin improvements in labour-intensive industries such as healthcare and logistics.

Importantly, our analysis indicates that human oversight and judgment remain essential. In healthcare, for example, robots could take on routine or physically demanding tasks, allowing professionals to focus on higher-value, patient-centric care. This collaborative approach would help mitigate demographic headwinds, lower healthcare costs and sustain service quality, positioning robots likely as a complement to human workforces rather than a substitute.

FIGURE 22
Where can humanoids work?

Sector	Jobs	Why task is more suited to humanoid
Manufacturing & Logistics	Assembly line work, packaging, material handling, warehouse sorting, loading/unloading	Tasks are repetitive, physically demanding or hazardous
Agriculture	Planting, harvesting, crop monitoring, livestock care	Labour shortages in rural areas and physically strenuous work
Healthcare & Elder Care	Patient lifting and transfer, basic caregiving, medication delivery, cleaning and sanitation	Roles are physically taxing and emotionally demanding
Construction	Heavy lifting, bricklaying, site cleanup, repetitive assembly tasks	High injury risk, physically demanding
Hospitality & Services	Room cleaning, dishwashing, luggage handling, repetitive food prep	Reluctance for low-status, physically demanding tasks

Source: Barclays Research



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